

Structure over Scale: Hierarchy-Constrained Modeling for English Online Polarization Detection

Mid-Submission Report

Team *Ugh*

International Institute of Information Technology, Hyderabad

{durga.nebhrajani, rakshitaggarwal.a, manav.berawal}@students.iiit.ac.in

Abstract

In SemEval-2026 Task 9, a text is polarized exactly when at least one polarization type applies to it. Most shared-task systems still predict detection and type separately and repair contradictions afterwards. We study whether this hierarchy should be built into the model. On the English data, four conditions share one DeBERTa-v3-base encoder: independent models (M1), a multi-task model (M2), post-hoc gating of M2 (M3), and M4-core, which derives detection from the type probabilities with a differentiable noisy-OR. We ran 100 training runs (5 folds $\times$ 5 seeds) and compared models with paired bootstrap tests. Unconstrained models contradict the hierarchy on 62% of texts; M3 and M4-core never do. M4-core does not improve the pre-declared metrics over M1 or M2 (RQ1), and it ties with post-hoc gating at equal consistency (RQ2). The analysis explains why. M4-core detects more polarized texts but its threshold calibration ignores neutral texts, so it raises false alarms. It also concentrates evidence in the dominant Political label and weakens rare labels. Gating, in turn, deletes 23% of correct type labels. These findings shape the plan for the final submission.

1 Introduction

Online polarization is language that divides people into opposing groups through hostility, stereotyping, or exclusion. Detecting it is hard because the signal is contextual: political words alone do not make a text polarized, and one text can attack several groups at once. SemEval-2026 Task 9 (POLAR) splits the problem into two related decisions (Naseem et al., 2026). **DET** (Subtask 1) asks whether a text is polarized. **TYPE** (Subtask 2) asks which of five dimensions it targets: Political, Racial/ethnic, Religious, Gender/sexual identity, or Other.

The two decisions are *nested*. Type labels exist only for polarized texts, so a text is polarized if and

only if at least one type applies. Standard multi-task models ignore this. They put separate heads on a shared encoder, and nothing stops them from predicting “not polarized” together with a type, or “polarized” with no type. Shared-task teams noticed these contradictions (Guo and Chang, 2026) and fixed them with rules applied after prediction (Lin et al., 2026; Maharjan et al., 2026). Maharjan et al. (2026) note that their hierarchical constraint “slightly limits raw multi-label recall”.

Our project asks whether the hierarchy should instead be part of the model. Our structured model, M4-core, has no separate detection head: the probability that a text is polarized is computed from the five type probabilities, so contradictions cannot occur. This report covers the first stage of the project, which answers two questions:

- **RQ1.** Does deriving DET from TYPE improve DET and TYPE macro-F1 over independent models and multi-task learning?
- **RQ2.** At equal consistency, does building the hierarchy into training keep more TYPE recall than fixing outputs afterwards?

Progress summary. We built a complete, tested pipeline and ran a controlled comparison: 100 training runs plus 75 derived conditions, with paired statistics. The answers are clear but not the ones we expected. RQ1 is *not supported*: M4-core is below both baselines on the pre-declared metrics. RQ2 is *inconclusive*: M4-core and post-hoc gating tie. Our analysis traces both outcomes to specific, fixable causes and to a limit that any consistent model must face. These findings shape the remaining work: a decoder extension, a new parent-to-child model (RQ5), and the two research questions kept from our proposal (§9). Our code is at https://github.com/Rakshitagg06/ANLP_Ugh, and all training runs are logged on Weights & Biases (§5).

2 Problem Statement

Task. Each input is an English text x . The gold labels are a detection label $d \in \{0, 1\}$ and a type vector $\mathbf{t} \in \{0, 1\}^5$. The gold data satisfy the hierarchy exactly:

$$d = 1 \iff \bigvee_{k=1}^5 t_k = 1. \quad (1)$$

A system outputs hard predictions $\hat{d}$ and $\hat{\mathbf{t}}$.

Consistency. A prediction is *consistent* if it satisfies Eq. 1. We measure violations with a symmetric version of the Logical Violation Rate (LVR). Maharjan et al. (2026) defined LVR for manifestations (Subtask 3) predicted on texts classified as non-polarized; we adapt it to the DET/TYPE pair and count both directions:

$$\text{LVR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{d}_i \neq \bigvee_k \hat{t}_{ik}], \quad (2)$$

split into **LVR-A** ($\hat{d} = 0$ but some $\hat{t}_k = 1$) and **LVR-B** ($\hat{d} = 1$ but every $\hat{t}_k = 0$).

Goal and research questions. We want models that are accurate *and* consistent. Before running any experiment, we fixed two hypotheses:

- **H1 (RQ1).** The structured model M4-core has higher DET macro-F1 and higher TYPE macro-F1 than both M1 and M2.
- **H2 (RQ2).** At equal consistency (LVR = 0), M4-core has higher TYPE macro-F1 and macro-recall than post-hoc gating (M3).

These six comparisons form one pre-declared test family.

Scope. This stage covers English Subtasks 1–2 and uses cross-validation on the official training set. Label-aware representations and the comparison with very large models, both planned in our proposal, move to the final stage.

3 Literature Review

The POLAR benchmark. SemEval-2026 Task 9 covers 22 languages and received submissions from 67 teams, with 69 system papers (Naseem et al., 2026). The English data have 3,222 training, 160 development and 1,452 test texts. Two facts from the task paper shaped our design. First, English annotator agreement is only moderate (Fleiss’ $\kappa = 0.39$), so some labels

are genuinely debatable. Second, the headroom differs between subtasks. The best English DET score is 0.825 against a baseline of 0.773, while the best TYPE score is 0.532 against 0.347. Detection is close to saturated; type classification is where improvement is most needed.

How teams solved the task. Large generative models dominated: the Qwen family was used by 31% of systems, LLaMA by 20%, Gemma/Gemini by 19%, and BERT-style encoders by only 7% (Naseem et al., 2026). The best English systems fine-tuned 12B–27B models (Tangkulung and Tsuruoka, 2026; Lin et al., 2026). Tangkulung and Tsuruoka (2026) predict all labels in a single forward pass, which they report reduces error propagation compared with multi-step decoding. Encoder systems remain competitive. Daniyal et al. (2026) fine-tune BERT-based models, and Trần (2026) use DeBERTa as a high-recall filter before an LLM assigns fine-grained labels. Bokaei et al. (2026) combine encoder ensembles with thresholds tuned on out-of-fold predictions. Orsanigo et al. (2026) use multi-task learning across the subtasks, and Nestor et al. (2026) add label-wise attention. This literature shows that a 184M-parameter encoder is a reasonable base for a controlled study, even if it is not the strongest system.

The hierarchy in POLAR systems. Several teams met the hierarchy, each with a different fix. Guo and Chang (2026) train the subtasks separately, observe that independent modelling “causes hierarchical violations”, and suggest post-hoc gating or hierarchical calibration as future work. Lin et al. (2026) repair one direction with a rule: if a text is predicted polarized but no type is predicted, they assign the Other label. Mohammad (2026) use a binary gatekeeper in front of specialist classifiers trained only on polarized content. Djamai et al. (2026) use their Subtask 1 encoder as a hard filter before type classification for Arabic. Iannielli et al. (2026) introduce a two-stage cascade to reduce false positives under class imbalance. The most explicit treatment is by Maharjan et al. (2026). They feed detached (stop-gradient) Subtask 1 and 2 predictions to the manifestation head, enforce $\hat{y}_{S1} = 0 \Rightarrow \hat{y}_{S2} = 0, \hat{y}_{S3} = 0$ at inference, and report violation rates as high as 33% for their ensemble. On English, their hierarchical variant scores 0.826 on DET and 0.506 on TYPE, against 0.752 and 0.511 for their non-hierarchical

variant. In every one of these systems the hierarchy acts *after* prediction, or between separately trained stages.

Multi-task learning. Sharing an encoder between related tasks can help both (Caruana, 1997), and the loss weights can be learned (Kendall et al., 2018). Sharing can also hurt: Guo and Chang (2026) report negative transfer when sparse fine-grained labels share a model with detection. Le and Phung (2026) found that initialising the type model from detection weights barely changed the result on Chinese (0.8312 with it, 0.8316 without). Sharing parameters alone does not appear to transfer the hierarchy.

Hierarchies and logical constraints. Outside POLAR, hierarchical classification is well studied. Silla and Freitas (2011) survey the field and describe how top-down systems propagate an error at a parent node to all its children, which is exactly the risk of post-hoc gating. Giunchiglia and Lukasiewicz (2020) build hierarchy consistency into the network and its loss rather than repairing outputs. Xu et al. (2018) derive a training loss from logical constraints, and Giunchiglia et al. (2022) survey where constraints can be placed: in the architecture, in the loss, or at inference. Our M4-core uses a noisy-OR, a classic model of several independent causes producing one effect (Pearl, 1988). Label-wise attention for multi-label text classification goes back to Mullenbach et al. (2018).

Thresholds and evaluation. Most POLAR systems tuned per-label thresholds (Naseem et al., 2026), and the F1-optimal threshold of a classifier depends on its calibration (Lipton et al., 2014). We therefore treat threshold tuning as part of each model. All models use the same TYPE threshold grid; M1 and M2 also tune a DET threshold, while M4-core has none. For statistics we follow the paired bootstrap of Koehn (2004), correct for multiple tests with Holm (1979), and build folds with multi-label stratification (Sechidis et al., 2011).

Gap and viability. No POLAR system makes the detection decision a *function* of the type predictions, and none compares post-hoc and training-time consistency under one controlled setup. That is the gap we address. The study is viable: the gold data satisfy the hierarchy exactly, the encoder is small enough to run 100 training runs in 3.9 GPU-

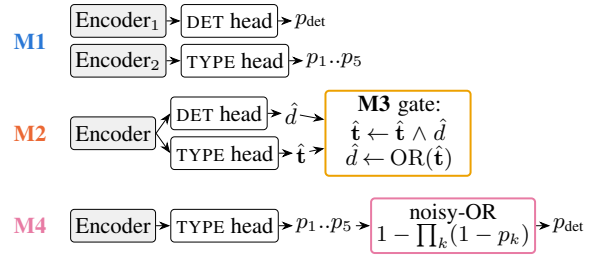


Figure 1: The four conditions. All use DeBERTa-v3-base with mean pooling and linear heads. M3 is not trained: it rewrites M2’s predictions. M4-core has no DET head; detection is computed from the type probabilities, so the hierarchy always holds.

hours (§5), and the shared-task literature supplies the baselines and metrics we need.

4 Models

All models use DeBERTa-v3-base (He et al., 2023) (184M parameters), mean pooling over tokens, dropout 0.1 and linear heads (Figure 1). Let p_k be the predicted probability of type k . Type labels are undefined for non-polarized texts, so the TYPE loss is a weighted binary cross-entropy (BCE) computed only on gold-polarized texts, with per-label positive weights.

M1 (independent). Two separately trained encoders, one for DET and one for TYPE, as in Guo and Chang (2026). Their outputs are joined without any consistency rule. M1 has 368M parameters.

M2 (shared multi-task). One encoder with a DET head and a TYPE head, trained with $\mathcal{L} = \text{BCE}_{\text{det}} + \lambda \mathcal{L}_{\text{type}}$ and $\lambda = 1$. Nothing links the two heads.

M3 (post-hoc gating). M3 is derived from M2’s predictions and reuses M2’s thresholds. It applies the inference rule of Maharjan et al. (2026) to the type labels: if $\hat{d} = 0$, all types are set to 0 (*one-way*). The *symmetric* variant then sets $\hat{d} \leftarrow \bigvee_k \hat{t}_k$, which guarantees LVR = 0.

M4-core (noisy-OR). The same encoder and TYPE head as M2, but no DET head. Detection is

$$p_{\text{det}} = 1 - \prod_{k=1}^5 (1 - p_k), \quad (3)$$

the probability that at least one type applies, computed in log space for numerical stability. It is trained with the same loss as M2. On a non-polarized text the DET loss pushes *every* type prob-

ability down, which supervises the type head exactly where the TYPE loss is masked. At inference, $\hat{t}_k = 1[p_k \geq \tau_k]$ and $\hat{d} = \bigvee_k \hat{t}_k$, so LVR = 0 by construction. The word “core” marks that it keeps M2’s head design, so M3 and M4-core differ only in where the hierarchy is enforced.

5 Experimental Setup

Data. All results use out-of-fold predictions on the 3,222 English training texts: 1,175 polarized and 2,047 not. Type support is Political 1,150, Racial/ethnic 281, Religious 112, Gender/sexual 72 and Other 126. Among the polarized texts, 426 have two or more types. Political appears in 98% of polarized texts, so it behaves almost like a copy of DET. The gold labels satisfy Eq. 1 with no exceptions. The supplied test file also contains labels. We used them only to check the hierarchy (533 polarized, 919 not, no violations); because their provenance relative to the official release is unconfirmed, no model is trained or scored on them. One training text also appears in the test file under a different id; it does not affect our cross-validation. The 160-text development set is too small to compare rare labels and was not used.

Protocol. We use one fixed five-fold split, stratified over all six labels (Sechidis et al., 2011), and five seeds (13, 21, 42, 87, 100), giving 4 trained conditions $\times$ 5 folds $\times$ 5 seeds = 100 training runs. In each run, 10% of the training folds form an inner validation set. Early stopping, the best epoch and all thresholds are chosen there; the held-out fold is predicted once. Training uses AdamW (Loshchilov and Hutter, 2019) with learning rate 2×10^{-5} , weight decay 0.01, 10% warm-up, effective batch size 16 and FP16, for at most 8 epochs with patience 2. Texts are truncated to 256 tokens.

Thresholds. Each type threshold τ_k is chosen from $\{0.20, 0.25, \dots, 0.80\}$ to maximize that label’s F1 on the inner-validation polarized texts. M1 and M2 also choose a DET threshold for DET macro-F1. M4-core has no DET threshold.

Metrics. DET macro-F1 averages the F1 of both classes. TYPE macro-F1, precision and recall average over the five labels and, as fixed before the experiments, are computed on *gold-polarized texts* only. We also report TYPE on *all* texts, where a type predicted for a non-polarized text counts as a false positive.

Statistics. For each seed, the five held-out folds form one prediction set over all 3,222 texts. We report the mean $\pm$ standard deviation over the five seeds. For comparisons we use a paired bootstrap over texts (2,000 resamples, Koehn, 2004): each resample applies the same texts to both models and all seeds. The six pre-declared tests are Holm-corrected (Holm, 1979). All other comparisons are exploratory.

Compute. All runs used one NVIDIA RTX 4060 laptop GPU. A run takes 2–3 minutes, and the full sweep took 3.9 GPU-hours.

Code and experiment logs. The code, configurations, saved analysis outputs and this report are available at:

- **Code (GitHub):**
https://github.com/Rakshitagg06/ANLP_Ugh
- **Weights & Biases:**
<https://wandb.ai/rakshitagg06-iiit-hyderabad/anlp-project>

The W&B project contains all 100 training runs, with their loss curves, configurations, and saved predictions and metrics.

6 Results

Table 1 gives the overall picture. M1 is the strongest model on every pre-declared metric. The two consistent models, M3 and M4-core, have lower TYPE scores on gold-polarized texts, and the gap comes from recall (about 0.47 vs. 0.60), not precision.

6.1 Consistency

The unconstrained models contradict the hierarchy on **62%** of all texts, almost entirely as LVR-A. M2 predicts at least one type for every one of the 2,047 non-polarized texts, and M1 for 2,039 of them. The reason is how their type heads are trained. The TYPE loss never sees a non-polarized text, positive weights push type probabilities up, and the thresholds are tuned only on polarized texts. Nothing teaches these heads to stay silent. M3 and M4-core have zero violations in every run.

6.2 RQ1: Does noisy-OR beat M1 and M2?

Table 2 and Figure 2 show that **H1 is not supported**. M4-core is below M1 on both metrics

Model	Params	DET	TYPE, gold-polarized texts				TYPE, all texts		LVR (%)	
		macro-F1	macro-F1	macro-P	macro-R		macro-F1		A	B
M1 (independent)	368M	0.799 $\pm$.007	0.543 $\pm$.013	0.509 $\pm$.024	0.600 $\pm$.023		0.299 $\pm$.013		61.9	0.0
M2 (shared MTL)	184M	0.793 $\pm$.003	0.497 $\pm$.015	0.444 $\pm$.018	0.595 $\pm$.019		0.255 $\pm$.029		61.5	0.0
M3 (gating)	184M	0.793 $\pm$.003	0.453 $\pm$.022	0.469 $\pm$.033	0.468 $\pm$.036		0.382 $\pm$.017		0.0	0.0
M4-core (noisy-OR)	184M	0.784 $\pm$.004	0.461 $\pm$.007	0.465 $\pm$.011	0.474 $\pm$.022		0.396 $\pm$.008		0.0	0.0

Table 1: **Main results** (out-of-fold over 3,222 texts; mean $\pm$ sd over five seeds; best in bold). M3 is the symmetric variant; the one-way variant has identical TYPE predictions and differs on DET for only two texts in one seed. LVR-A: “not polarized” but a type is predicted. LVR-B: “polarized” but no type is predicted.

Comparison	Δ	95% CI	p_{Holm}
RQ1	M4–M1 DET F1	−0.014 [−0.023, −0.006]	<0.001
	M4–M1 TYPE F1	−0.082 [−0.102, −0.064]	<0.001
	M4–M2 DET F1	−0.008 [−0.016, −0.001]	0.072
	M4–M2 TYPE F1	−0.036 [−0.055, −0.019]	0.004
RQ2	M4–M3 TYPE F1	+0.008 [−0.009, +0.022]	0.732
	M4–M3 TYPE R	+0.006 [−0.014, +0.023]	0.732

Table 2: **Pre-declared tests.** Δ is the seed-averaged difference; the CI is a 95% paired bootstrap interval. TYPE scores use gold-polarized texts.

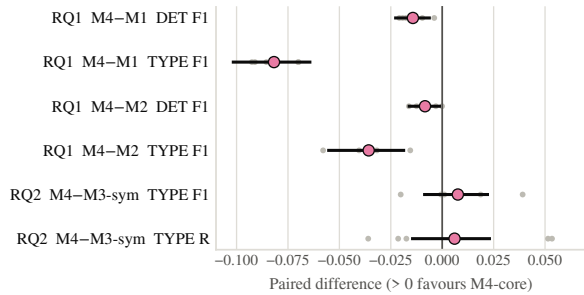


Figure 2: **The six pre-declared tests.** Pink: mean difference (M4-core minus baseline). Black bar: 95% CI. Grey: the five per-seed differences. All RQ1 bars lie left of zero (M4-core worse); both RQ2 bars cross zero (no detectable difference).

and below M2 on TYPE; these three differences remain significant after Holm correction, and no seed favours M4-core. Against M2 on DET, M4-core is lower in all five seeds, but the difference is not significant after correction ($p = 0.072$). Two mechanisms explain the result.

The DET gap comes from false alarms. M4-core finds more polarized texts but also flags more neutral ones (Table 3). Its DET recall rises from 0.761 (M2) to 0.831, while precision falls from 0.722 to 0.674. The cause is the decision rule. M4-core says “polarized” when *any* type passes its threshold, and those thresholds were tuned only on polarized texts, so their tuning never saw the cost of a false alarm. The Political threshold, which

Model	DET P	DET R	FP	FN	Neutral w/ type
M1	0.733	0.761	325	280	2,039
M2	0.722	0.761	345	281	2,047
M3	0.722	0.761	345	281	345
M4	0.674	0.831	472	199	472

Table 3: **Detection errors** (per seed, 3,222 texts). P/R: precision and recall of the polarized class. FP: neutral texts flagged as polarized. FN: polarized texts missed. Last column: neutral texts given at least one type.

almost decides detection on its own, fell to the bottom of the grid (0.20) in 19 of 25 runs. M1 and M2, by contrast, tune a dedicated DET threshold for DET macro-F1. The DET comparison is therefore not calibration-matched.

The TYPE gap is the price of consistency. Under the pre-declared metric, a type predicted for a neutral text is not counted as an error. M1 and M2 benefit fully from this. A consistent model cannot: on a polarized text it can only output a type if it also detects the text, so its type recall is capped by its detection. The RQ1 TYPE test therefore compares a consistent model with inconsistent ones. When the same predictions are scored on all texts, the ranking reverses (Table 1): M4-core scores 0.396, against 0.299 for M1 and 0.255 for M2. The paired gains are +0.097 [+0.078, +0.114] over M1 and +0.141 [+0.121, +0.158] over M2, in all five seeds. This view is exploratory and does not change the pre-declared verdict, but it shows that the RQ1 outcome depends on what the metric chooses to count. We have not yet verified which scope the official SemEval scorer uses.

6.3 RQ2: Gating versus training-time structure

Both M3 and M4-core have zero violations, so RQ2 compares them at equal consistency. Their pre-declared scores are statistically tied (Table 2). The TYPE F1 difference is +0.008 with a 95% CI

	M3	M4-core	M4-M3 [95% CI]
Coverage	0.768	0.843	+0.075 [+0.064, +0.087]
Cond. recall	0.853	0.823	-0.030 [-0.041, -0.020]
Micro recall	0.655	0.694	+0.038 [+0.027, +0.049]
Macro recall	0.468	0.474	+0.006 [-0.014, +0.023]
<i>Median p_k on texts that carry label k (M2 $\rightarrow$ M4-core)</i>			
Political	0.577 $\rightarrow$ 0.754	Religious	0.772 $\rightarrow$ 0.419
Gender	0.354 $\rightarrow$ 0.051	Other	0.392 $\rightarrow$ 0.159

Table 4: **Why RQ2 ties** (exploratory). Coverage: share of gold type labels on texts the model detects. Conditional recall: share of those labels it then predicts. Micro recall = coverage $\times$ conditional recall. The first three differences hold in all five seeds.

Label	n	Δ recall [95% CI]	Δ F1 [95% CI]
Political	1,150	+0.067 [+0.055, +0.078]	+0.040 [+0.033, +0.048]
Racial/ethnic	281	-0.011 [-0.041, +0.019]	-0.026 [-0.051, -0.003]
Religious	112	-0.016 [-0.059, +0.026]	-0.006 [-0.041, +0.030]
Gender/sexual	72	+0.064 [+0.009, +0.116]	+0.039 [-0.014, +0.089]
Other	126	-0.073 [-0.113, -0.034]	-0.009 [-0.039, +0.019]

Table 5: **RQ2 per label**: M4-core minus M3 on gold-polarized texts (exploratory). Bold: the 95% interval excludes zero.

of $[-0.009, +0.022]$, and the seeds split 3/5 and 2/5. **H2 is not supported**, and neither is the opposite. The intervals do rule out large effects: the upper end of the recall interval, +0.023, is less than a fifth of the 0.127 recall that gating costs relative to M2.

Two opposite effects cancel. For a consistent model, label recall equals *coverage* (the share of gold labels on texts it detects) times *conditional recall* (how many of those labels it then predicts). Table 4 shows that the models sit at opposite ends of this product. M3 keeps M2’s discrimination (0.853) but inherits the 23% of labels on texts M2’s detector misses. M4-core detects more (coverage +0.075) but discriminates less well among detected texts (-0.030). Both effects are consistent across all seeds. The coverage gain falls mainly on Political, which carries two thirds of all type labels. The discrimination loss falls on Racial/ethnic, Religious and Other. Micro recall, which weights labels by frequency, therefore favours M4-core (+0.038), while macro recall, which weights all labels equally, shows a tie.

The tie hides per-label differences. Table 5 shows the same pattern label by label. M4-core is clearly better on Political and recovers more of the rarest label, Gender/sexual. It is worse on Other and less precise on Racial/ethnic. A macro aver-

Proposal item	Status
Stratified 5-fold CV, metrics, LVR	Done
M1 and M2 (5 folds $\times$ 5 seeds)	Done
M3 gating	Done (re-implemented)
M4 noisy-OR	Done as M4-core
Full 5 $\times$ 5 ladder and paired tests	Done (ahead of plan)
Label-aware attention, Other head	Kept as RQ3
TF-IDF, BERT, RoBERTa baselines	Deferred
Learned loss weighting	Deferred
Comparison with 12B–27B systems	Kept as RQ4
Leaderboard claim	Not made (RQ4)

Table 6: Status of the work planned in the proposal.

age gives each label equal weight, so these gains and losses cancel.

Why M4-core weakens rare labels. The noisy-OR loss does two things at once (Table 4, bottom). On neutral texts it pushes every type probability close to zero (median ≤ 0.012 , against up to 0.55 for M2), which is the intended effect. But on polarized texts, the detection target can be satisfied by Political alone, so the loss rewards raising Political and gives little reason to raise the other heads. As a result, rare labels are suppressed even on texts that truly have them: the median Gender probability on Gender texts falls from 0.354 to 0.051. This is a credit-assignment problem. In Eq. 3, once one head is confident, the gradient reaching the other heads becomes small.

What gating removes. Per seed, gating deletes 4,379 of the 6,647 type labels M2 predicts. 87.7% of them are false alarms on neutral texts, which is gating working as intended. But it also deletes **338 correct labels, 22.9%** of all correct type predictions M2 makes. Every one of them is on a polarized text that M2’s detection head missed, and on 98.7% of those texts M2 had already predicted at least one correct type. This is the error propagation that Silla and Freitas (2011) describe for top-down classifiers. The loss is larger on single-label texts (-0.239 recall) than on multi-label texts (-0.161). The cost of gating therefore comes from detection misses, not from the number of labels on a text.

7 Progress Against the Proposal

Table 6 compares our progress with the proposal. The controlled four-model comparison, which the proposal scheduled for mid-October, is complete, including the full five-fold, five-seed sweep and the significance tests. Three changes were delib-

erate:

- **M4 became M4-core.** The proposal’s M4 combined label-aware attention with noisy-OR. Testing both at once would make it impossible to tell which one causes a difference, so we first isolated the hierarchy mechanism. Label-aware attention remains RQ3.
- **M3 was re-implemented.** Instead of adapting the released Sagarmatha system, we applied its inference rule to our own M2, so that M3 and M4-core share the same encoder, data and thresholds.
- **Classical baselines were deferred.** The TF-IDF, BERT and RoBERTa baselines do not affect RQ1 or RQ2, so they come after the controlled comparison.

We also built the infrastructure: configuration-driven training, saved predictions, experiment tracking, an analysis pipeline that regenerates every table, and unit tests, which caught two bugs before the main sweep.

8 Discussion

We expected the structured model to avoid gating’s recall loss because it “discards nothing”. This was too simple: a consistent model cannot recover a type on a text it fails to detect, and where the constraint is enforced only changes who pays for that limit. Post-hoc gating keeps the unconstrained model’s type discrimination but inherits its detector’s misses. Training-time noisy-OR detects more texts, but its loss concentrates evidence in the dominant label and its decoder inherits thresholds that never saw neutral text.

Three lessons follow. First, the evaluation scope matters: whether TYPE is scored on polarized texts or on all texts reverses the model ranking. Second, calibration is part of the model. M4-core’s main weaknesses come from how its probabilities are turned into decisions, not only from how it is trained. Third, the direction of the hierarchy matters. Deriving the parent from the children lets a confident child switch on the parent, and lets it absorb the credit that should reach the other children. These lessons define the next stage.

9 Remaining Work and Research Questions

RQ1 and RQ2 are closed under the frozen protocol, and their answers stay as reported. The open

question is no longer whether noisy-OR wins on the frozen macro-F1 numbers. It is whether the consistency–recall trade-off comes from the *decoder*, from *aggregating* types into detection, or from the *hierarchy itself*. The remaining work tests these in that order. It keeps the two research questions from our proposal (RQ3 and RQ4) and adds one new question (RQ5). All primary tests and margins will be fixed before any new run.

Decoder extension (not a new RQ). *Does M4-core improve once its thresholds see neutral texts?* In the current code, TYPE thresholds use only gold-polarized inner-validation texts, and no inner-validation probabilities or checkpoints were saved, so a proper test needs a code change and new runs. We will rerun M4-core (25 runs) with thresholds tuned on *all* inner-validation texts over the grid $\{0.05, 0.10, \dots, 0.80\}$, and keep checkpoint selection on gold-polarized TYPE macro-F1, so that only the threshold rule changes. Because M3 is derived from M2, we will also rerun M2 (25 runs). Both reruns save inner-validation probabilities, which lets us place M3 and M4-core at the same detection precision, and separately at the same detection recall, without tuning on held-out folds. Re-thresholding the already saved held-out predictions would tune on test data, so we will report it only as exploratory. *Why it matters:* RQ1 traced M4-core’s DET loss to thresholds tuned without neutral texts, and RQ2 compared the two consistent models at different operating points. This is the cheapest test of whether those results are decoding artefacts. The results will be reported beside the frozen RQ1/RQ2 tables, not instead of them.

RQ5 (new): Does the direction of the hierarchy matter? *Can a calibrated parent-to-child model improve TYPE while keeping DET as good as M2’s?* M4-core builds detection from the types. The new model, M5, does the reverse. It keeps an independent DET head d and a conditional type head q_k , and predicts $p_k = d \cdot q_k$, the factorization $P(t_k | x) = P(d | x) P(t_k | d, x)$ implied by the hierarchy. Low detection confidence then softly lowers every type, and a confident type can no longer switch detection on. The DET head is trained with BCE on all texts. Because q_k means $P(t_k | d = 1, x)$, its weighted BCE is computed on gold-polarized texts only. Thresholds are tuned on all inner-validation texts over the same grid as the decoder extension. We will test two

decoders with the same encoder, folds and seeds: *soft*, $\hat{t}_k = \mathbf{1}[dq_k \geq \tau_k]$ and $\hat{d} = \mathbf{1}[d \geq \tau_D]$; and *hard*, which also requires $\hat{d} = 1$ for any $\hat{t}_k = 1$. Two hypotheses are fixed in advance. **H5a:** M5 has higher TYPE macro-F1 than M2 and than the decoder-extension M4-core, on the scoring scope fixed on Oct 2–3. **H5b** (non-inferiority): DET macro-F1 is at most $\delta = 0.01$ below M2. H5b is decided by the lower end of the 95% paired bootstrap interval for M5–M2, which must lie above -0.01 ; the point estimate alone is not enough. *Why it matters:* the decoder extension can fix M4-core’s thresholds, but not the fact that one confident type switches detection on and absorbs the credit for the others (§6.3). RQ5 tests whether the hierarchy helps when it follows the direction of the label definition.

RQ3 (proposal): Do label-aware representations help rare labels? *Do label-specific representations improve rare-label and multi-label TYPE performance, especially for Other?* As in our proposal, each named type gets a learned query that attends over the tokens (Mullenbach et al., 2018; Nestor et al., 2026), and Other gets a separate head on the pooled representation, since it has no single meaning. RQ3 will run on one hierarchy variant, chosen by a fixed keep-rule: the M5 variant that passes H5b; if both pass, the one with higher TYPE macro-F1; if neither passes, the decoder-extension M4-core. This way a new head is not confounded with a new constraint. *Why it matters:* Gender/sexual and Other are the weakest labels for every model (F1 below 0.30) and lose most under noisy-OR, and all five types currently share one pooled vector. Type classification is also where the task has the most headroom (0.532 best vs. 0.347 baseline).

RQ4 (proposal): How does a 184M model compare with 12B–27B systems? *How does our best structured encoder compare with the much larger systems that lead the English leaderboard, relative to model size?* We will place our out-of-fold scores beside the published English results (0.825 DET, 0.532 TYPE) and state clearly that the two come from different data splits. We will not call this a leaderboard comparison. We will compare two models: the variant chosen by the keep-rule, trained once more with its checkpoint kept, and the frozen M4-core. The development set will be scored once, as a sanity check, and will not be used to choose between models. The test file stays

Dates	Work
Oct 2–3	Read the official Subtask 2 scorer and record whether later TYPE scores use all texts or gold-polarized texts only.
Oct 4–6	Decoder extension: change threshold selection to all inner-validation texts (grid 0.05–0.80) and save inner-validation probabilities; rerun M4-core and M2 (25 runs each); matched operating-point comparison of M3 and M4-core.
Oct 7–16	RQ5: parent-to-child M5, soft and hard decoders (25 runs each); test H5a and H5b; apply the keep-rule.
Oct 17–23	RQ3: label-aware heads and a separate Other head on the kept variant; single- vs. multi-label analysis.
Oct 24–27	RQ4: kept variant (retrained with checkpoint) and frozen M4-core vs. published 12B–27B results; development set scored once.
Oct 28–31	Error analysis of Other and neutral false alarms; final report and code release.

Table 7: Plan to the final submission (October 31). RQ1 and RQ2 are closed; every later row is a new experiment.

unused until its labels are confirmed; the one text it shares with the training set will then be left out. *Why it matters:* it measures how much of the gap to large models a small structured encoder closes, the “structure over scale” question.

Deferred. The TF-IDF, BERT and RoBERTa baselines and learned loss weighting (Kendall et al., 2018) are deferred in favour of the decoder extension and RQ5.

10 Timeline

Table 7 gives the plan. Each 25-run condition takes about one GPU-hour on our hardware, so the training budget is not a risk. The main risks are scientific: M5 may lose DET recall through its gate and fail H5b, in which case RQ3 falls back to the decoder-extension M4-core; and if the test-label question stays open, RQ4 will say so and rely on the published numbers.

11 Conclusion

Unconstrained models contradict the POLAR hierarchy on most texts. A noisy-OR model removes every contradiction but does not beat independent or multi-task models on our pre-declared metrics, and it ties with post-hoc gating, which itself removes almost a quarter of correct labels. We traced these outcomes to specific, testable causes, and the final stage tests them directly.

Limitations

The pre-declared TYPE metric scores only gold-polarized texts, which favours unconstrained models, and we have not yet confirmed the official scorer’s scope. Detection decisions are not calibration-matched across models, and M1 has twice the parameters of the others. Bootstrap intervals capture variation over texts; training variation is reflected only through five seeds, and only the six pre-declared tests are corrected for multiple comparisons. The decomposition and probability analyses were designed after seeing the main results. All results are English cross-validation scores from one consumer GPU, so they are not leaderboard-comparable.

References

- Zahra Bokaei, Alessandra Terranova, Yi Zheng, Tom Bidewell, and Bjorn Ross. 2026. [SMASH at SemEval-2026 task 9: Detecting multilingual polarisation with encoder ensembles and calibrated decision thresholds](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 832–848, San Diego, California, USA. Association for Computational Linguistics.
- Rich Caruana. 1997. Multitask learning. *Machine Learning*, 28(1):41–75.
- Syeda Samah Daniyal, Muneeba Badar, Manal Hasan, Shifa Shah, Sandesh Kumar, and Abdul Samad. 2026. [MSqrd at SemEval-2026 task 9: Detecting multilingual, multicultural and multievent online polarization](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 1664–1671, San Diego, California, USA. Association for Computational Linguistics.
- Abdelbasset Djamai, Sahara Al-Madi, Norah Al-Zaid, Khlood Al Jallad, and Mona Azim. 2026. [NA-MAA at SemEval-2026 task 9: Comparing generative, retrieval-augmented, and discriminative methods for Arabic online polarization detection and type classification](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 3562–3590, San Diego, California, USA. Association for Computational Linguistics.
- Eleonora Giunchiglia and Thomas Lukasiewicz. 2020. Coherent hierarchical multi-label classification networks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9662–9673.
- Eleonora Giunchiglia, Mihaela Catalina Stoian, and Thomas Lukasiewicz. 2022. [Deep learning with logical constraints](#). In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI-22*.
- Fengze Guo and Yue Chang. 2026. [YEZE at SemEval-2026 task 9: Detecting multilingual, multicultural and multievent online polarization via heterogeneous ensembling](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 1860–1873, San Diego, California, USA. Association for Computational Linguistics.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations*.
- Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Angelo Iannielli, Samuele Maroli, Marco Roberto, Stefano Sammartino, Valentino Vacirca, Claudio Savelli, Riccardo Coppola, and Flavio Giobergia. 2026. [MINDS at SemEval-2026 task 9: A multi-paradigm approach to cross-lingual polarization detection](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 2475–2486, San Diego, California, USA. Association for Computational Linguistics.
- Alex Kendall, Yarin Gal, and Roberto Cipolla. 2018. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 7482–7491.
- Philipp Koehn. 2004. [Statistical significance tests for machine translation evaluation](#). In *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 388–395, Barcelona, Spain. Association for Computational Linguistics.
- Minh-Hoang Le and Khoan Phung. 2026. [AlphaLyræ at SemEval-2026 task 9: Metric learning and asymmetric loss for Chinese polarization analysis](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 503–508, San Diego, California, USA. Association for Computational Linguistics.
- Ding-Xiang Lin, Po-Chun Chu, and Lung-Hao Lee. 2026. [NYCU-NLP at SemEval-2026 task 9: Stacking small language models for multilingual, multicultural and multievent polarization detection](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 1258–1267, San Diego, California, USA. Association for Computational Linguistics.
- Zachary C. Lipton, Charles Elkan, and Balakrishnan Narayanaswamy. 2014. Optimal thresholding of classifiers to maximize F1 measure. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*.

- Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations*.
- Sujal Maharjan, Astha Shrestha, and Pratikshya Shrestha. 2026. [Sagarmatha at SemEval-2026 task 9: Heterogeneous ensembling and hierarchical task conditioning for multilingual latent distributional divergence modeling](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 3045–3051, San Diego, California, USA. Association for Computational Linguistics.
- Abdullah Mohammad. 2026. [PolaFusion at SemEval-2026 task 9: Ensemble transformers with targeted augmentation for multilingual polarization detection](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 2877–2885, San Diego, California, USA. Association for Computational Linguistics.
- James Mullenbach, Sarah Wiegrefe, Jon Duke, Jimeng Sun, and Jacob Eisenstein. 2018. [Explainable prediction of medical codes from clinical text](#). In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1101–1111, New Orleans, Louisiana. Association for Computational Linguistics.
- Usman Naseem, Robert Geislinger, Ada Ren, Sarah Kohail, Rudy Garrido Veliz, P Sam Sahil, Yiran Zhang, Marco Antonio Stranisci, Idris Abdulmumin, Özge Alacam, Cengiz Acarturk, Aisha Jabr, Saba Anwar, Abinew Ali Ayele, Elena Tutubalina, Aung Kyaw Htet, Xintong Wang, Surendrabikram Thapa, Tanmoy Chakraborty, and 15 others. 2026. [SemEval-2026 task 9: Detecting multilingual, multicultural and multievent online polarization](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 3779–3796, San Diego, California, USA. Association for Computational Linguistics.
- Maria Nestor, Maroan Al Shrafat, Ioana Pește, Daniela Gifu, and Diana Trandabăț. 2026. [PolarizedTeam at SemEval-2026 task 9: Detecting multilingual, multicultural and multievent online polarization](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 1391–1397, San Diego, California, USA. Association for Computational Linguistics.
- Veronica Orsanigo, Alan Ramponi, and Elisa Leonardelli. 2026. [DigiS-FBK at SemEval-2026 task 9: Multi-task learning for multilingual and cross-cultural polarization classification](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 2838–2851, San Diego, California, USA. Association for Computational Linguistics.
- Judea Pearl. 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA.
- Konstantinos Sechidis, Grigorios Tsoumakas, and Ioannis Vlahavas. 2011. On the stratification of multi-label data. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, pages 145–158.
- Carlos N. Silla and Alex A. Freitas. 2011. A survey of hierarchical classification across different application domains. *Data Mining and Knowledge Discovery*, 22(1–2):31–72.
- Howard Tangkulung and Yoshimasa Tsuruoka. 2026. [UTokyo tsuruoka lab at SemEval-2026 task 9: Efficient single forward pass inference for multi-label polarization classification](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 1641–1651, San Diego, California, USA. Association for Computational Linguistics.
- Hoàn Trần. 2026. [UIT-polar at SemEval-2026 task 9 detecting multilingual, multicultural and multievent online polarization](#). In *Proceedings of the 20th International Workshop on Semantic Evaluation (2026)*, pages 127–132, San Diego, California, USA. Association for Computational Linguistics.
- Jingyi Xu, Zilu Zhang, Tal Friedman, Yitao Liang, and Guy Van den Broeck. 2018. A semantic loss function for deep learning with symbolic knowledge. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*.